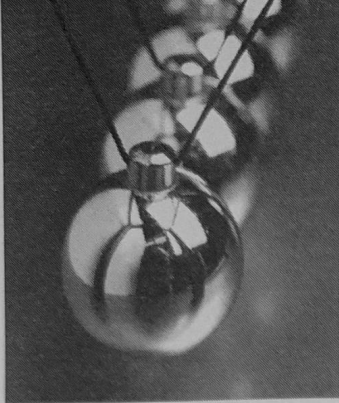




CHAPTER

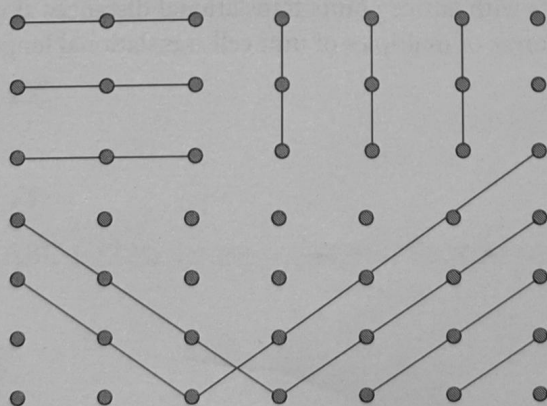
3

Crystal Planes, X-ray Diffraction and Defects in Solids

3.1 Crystal planes, directions and Miller indices

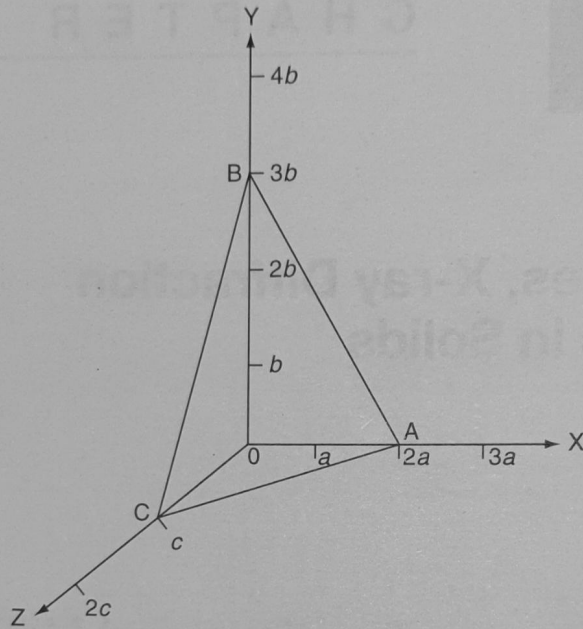
Crystal planes are defined as some imaginary planes inside a crystal in which large concentration of atoms are present. Inside the crystal, there exists certain directions along which large concentration of atoms exists. These directions are called crystal directions. Figure 3.1 shows a two-dimensional lattice with different orientations of crystal planes.

Figure 3.1 A two-dimensional lattice with crystal planes



Crystal planes and directions can be represented by a set of three small integers called Miller indices [because Miller derived a method of representing crystal planes]. These integers are represented in general as h , k and l . If these integers are enclosed in round brackets as (hkl) , then it represents a

Figure 3.2 Miller indices for a plane ABC



plane. On the other hand, if they are enclosed in square brackets as $[hkl]$, then it represents crystal direction perpendicular to the above-said plane. Next, we will see the way of obtaining Miller indices for a plane.

- (i) As shown in Fig. 3.2, take a lattice point as origin '0' of crystallographic axes X, Y and Z in a space lattice. The unit cell translational distances or lattice constants along X, Y and Z directions are a , b and c , respectively. Let a crystal plane ABC intersect these three axes at $2a$, $3b$ and c . In general, the intercepts can be represented as pa , qb , and rc .
- (ii) Divide these intercepts with lattice points translational distances along the axes to obtain intercepts of the plane in terms of multiples of unit cell translational lengths.

$$\text{i.e., } \frac{2a}{a}, \frac{3b}{b}, \frac{c}{c}$$

$$2, 3, 1$$

$$\text{in general } \frac{pa}{a}, \frac{qb}{b}, \frac{rc}{c}$$

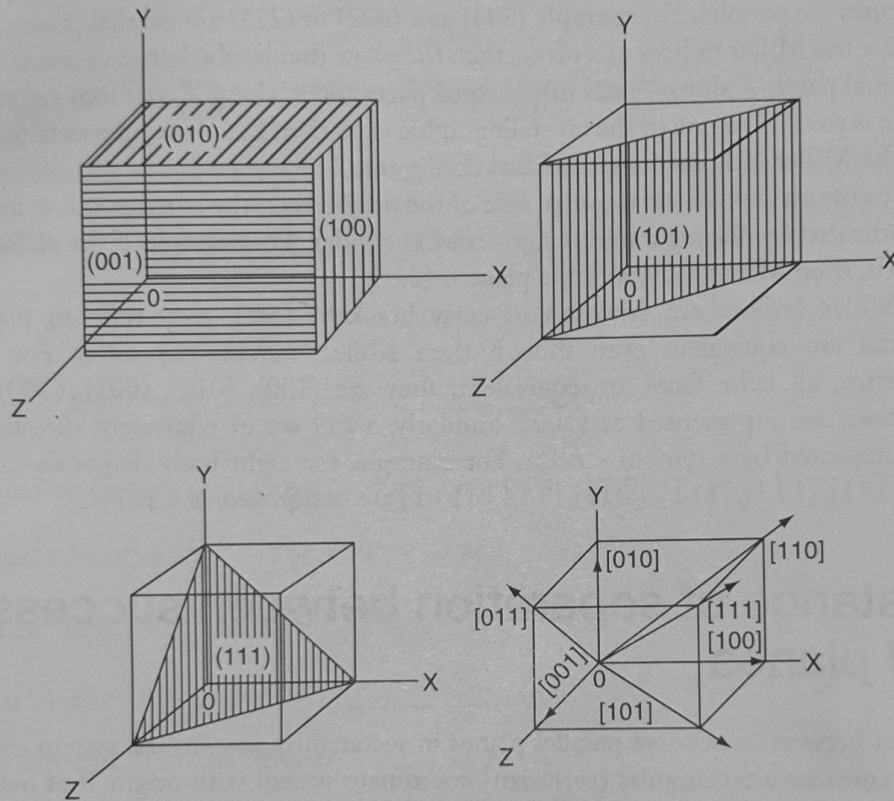
$$p, q, r$$

- (iii) Take the reciprocals of these multiples, they are $\frac{1}{2}, \frac{1}{3}, \frac{1}{1}$; in general $\frac{1}{p}, \frac{1}{q}, \frac{1}{r}$

- (iv) Clear these fractions [by multiplying with LCM] to smallest integers having the same ratio as the fractions, enclose these integers in brackets.

Figure 3.3

Represent some important crystal planes and directions in a cubic crystal



$$\frac{1}{2} \times 6, \frac{1}{3} \times 6, \frac{1}{1} \times 6$$

$$3 \quad 2 \quad 6$$

in general $\frac{pqr}{p}, \frac{pqr}{q}, \frac{pqr}{r}$

$$qr \quad pr \quad pq$$

Miller indices of the plane ABC is (326). In general, indices of a plane are represented as $(hkl) = (qr \text{ pr } pq)$

$$\text{or } \frac{1}{p} : \frac{1}{q} : \frac{1}{r} = h : k : l$$

Miller indices may be defined as a set of three small integers obtained by clearing the reciprocals of the three intercepts [in terms of multiples of unit cell edges] made by a plane on crystallographic axes.

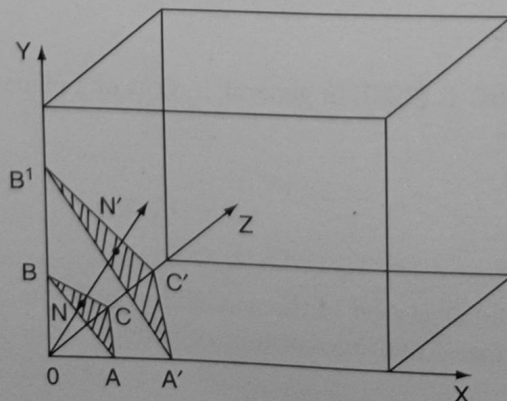
Now, we will see the important features of Miller indices:

- (i) Miller indices represent a set of equidistant parallel planes.
- (ii) If the Miller indices of a plane represent some multiples of Miller indices of another plane, then these planes are parallel. For example (844) and (422) or (211) are parallel planes.
- (iii) If (hkl) are the Miller indices of a plane, then the plane divides the lattice constant ' a ' along X-axis into h equal parts, ' b ' along Y-axis into k equal parts and ' c ' along Z-axis into l equal parts.
- (iv) If a plane is parallel to one of the crystallographic axes, then the plane intersects that axis at infinity and the Miller indices along that direction is zero.
- (v) If a plane cuts an axis on the negative side of the origin, then the corresponding index is negative and is indicated by placing a minus sign above the index. For example, if the plane cuts on negative Y-axis, then Miller indices of the plane is $(h\bar{k}l)$.
- (vi) When Miller indices are enclosed in curly brackets, $\{hkl\}$, they refer to planes which in the crystal are equivalent even though their Miller indices may differ. For example in cubic lattice, all cube faces are equivalent, they are (100), (010), (001), $(\bar{1}00)$, $(0\bar{1}0)$, $(00\bar{1})$. These planes are represented as $\{100\}$. Similarly, a full set of equivalent directions in a crystal is represented by a symbol $\langle hkl \rangle$. For example, the eight body diagonals of a cube $[111]$, $[\bar{1}\bar{1}\bar{1}]$, $[\bar{1}11]$, $[1\bar{1}1]$, $[11\bar{1}]$, $[\bar{1}\bar{1}1]$, $[1\bar{1}\bar{1}]$, $[\bar{1}1\bar{1}]$ are designated as $\langle 111 \rangle$.

3.2 Distance of separation between successive hkl planes

The separation between successive parallel planes in rectangular axes crystal system can be extracted easily. Let us consider a rectangular [cartesian] coordinate system with origin 'O' at one of the lattice points. Let (hkl) be the Miller indices of a plane ABC, which makes intercepts OA, OB and OC on X, Y and Z axes, respectively as shown Fig 3.4. A normal to this plane from the origin passes through a point N in the plane ABC, such that $ON = d_1$. This normal makes α' , β' , and γ' angles with X, Y and Z-axes, respectively. Since the plane segments ' a ' into ' h ' equal parts, ' b ' into ' k ' equal parts and ' c ' into ' l ' equal parts, then the intercepts OA, OB and OC are such that:

Figure 3.4 Orthorhombic crystal



$$OA = \frac{a}{b}, OB = \frac{b}{k} \quad \text{and} \quad OC = \frac{c}{l} \quad (3.1)$$

where a, b, c are the unit cell edge lengths along X, Y and Z-axes, respectively.

From Fig. 3.4

$$\cos \alpha' = \frac{d_1}{OA}, \cos \beta' = \frac{d_1}{OB} \quad \text{and} \quad \cos \gamma' = \frac{d_1}{OC} \quad (3.2)$$

Let the coordinates of N be x, y and z along X, Y and Z axes, then:

$$(ON)^2 = d_1^2 = x^2 + y^2 + z^2 \quad (3.3)$$

Also from Fig. 3.4:

$$\cos \alpha' = \frac{x}{d_1}, \cos \beta' = \frac{y}{d_1}, \cos \gamma' = \frac{z}{d_1} \quad (3.4)$$

Substitute Equation (3.4) in (3.3) gives:

$$\begin{aligned} d_1^2 &= d_1^2 \cos^2 \alpha' + d_1^2 \cos^2 \beta' + d_1^2 \cos^2 \gamma' \\ &= d_1^2 [\cos^2 \alpha' + \cos^2 \beta' + \cos^2 \gamma'] \end{aligned}$$

$$(\text{or}) \quad \cos^2 \alpha' + \cos^2 \beta' + \cos^2 \gamma' = 1 \quad (3.5)$$

Substitute Equation (3.2) in (3.5) gives:

$$\frac{d_1^2}{(OA)^2} + \frac{d_1^2}{(OB)^2} + \frac{d_1^2}{(OC)^2} = 1 \quad (3.6)$$

Again substitute Equation (3.1) in (3.6)

$$\frac{d_1^2 b^2}{a^2} + \frac{d_1^2 k^2}{b^2} + \frac{d_1^2 l^2}{c^2} = 1 \quad (\text{or}) \quad d_1^2 \left[\frac{b^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2} \right] = 1$$

$$d_1 = \frac{1}{\sqrt{\frac{b^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}}} \quad (3.7)$$

Let $\left(\frac{h}{2} \frac{k}{2} \frac{l}{2}\right)$ be the Miller indices of the next plane $A' B' C'$, this plane makes intercepts OA', OB' and OC' on X, Y and Z axes, respectively. A normal from the origin to this plane passes through a

point N' , so that $ON' = d_2$. As the extension of d_1 is d_2 , it makes same angles α' , β' and γ' with X, Y and Z-axes, respectively. Since the plane segments 'a' into $b/2$ equal parts, b into $k/2$ equal parts and into $l/2$ equal parts, then the intercepts OA' , OB' and OC' are such that:

$$OA' = \frac{a}{(b/2)} = \frac{2a}{b}, \quad OB' = \frac{b}{(k/2)} = \frac{2b}{k} \quad \text{and} \quad OC' = \frac{c}{(l/2)} = \frac{2c}{l} \quad (3.1)$$

From Fig. 3.4,

$$\cos \alpha' = \frac{d_2}{(OA')}, \quad \cos \beta' = \frac{d_2}{(OB')} \quad \text{and} \quad \cos \gamma' = \frac{d_2}{(OC')} \quad (3.2')$$

Let the coordinates of N' are x' , y' and z' along X, Y and Z-axes, respectively.

$$(ON')^2 = d_2^2 = x'^2 + y'^2 + z'^2 \quad (3.3')$$

Also from Fig 3.4:

$$\cos \alpha' = \frac{x'}{d_2}, \quad \cos \beta' = \frac{y'}{d_2} \quad \text{and} \quad \cos \gamma' = \frac{z'}{d_2} \quad (3.4')$$

Substitute Equation (3.4') in (3.3') gives:

$$\begin{aligned} d_2^2 &= d_2^2 \cos^2 \alpha' + d_2^2 \cos^2 \beta' + d_2^2 \cos^2 \gamma' \\ &= d_2^2 [\cos^2 \alpha' + \cos^2 \beta' + \cos^2 \gamma'] \end{aligned}$$

$$(\text{or}) \quad \cos^2 \alpha' + \cos^2 \beta' + \cos^2 \gamma' = 1 \quad (3.5')$$

Substitute Equation (3.2') in (3.5') gives:

$$\frac{d_2^2}{(OA')^2} + \frac{d_2^2}{(OB')^2} + \frac{d_2^2}{(OC')^2} = 1 \quad (3.6')$$

Again substitute Equation (3.1') in (3.6') gives:

$$\frac{d_2^2 b^2}{(2a)^2} + \frac{d_2^2 k^2}{(2b)^2} + \frac{d_2^2 l^2}{(2c)^2} = 1 \quad (\text{or}) \quad d_2^2 \left[\frac{b^2}{(2a)^2} + \frac{k^2}{(2b)^2} + \frac{l^2}{(2c)^2} \right] = 1$$

$$(\text{or}) \quad d_2 = \frac{1}{\sqrt{\frac{b^2}{(2a)^2} + \frac{k^2}{(2b)^2} + \frac{l^2}{(2c)^2}}} \quad (\text{or}) \quad d_2 = \frac{2}{\sqrt{\frac{b^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}}} \quad (3.7')$$

Let the separation between the planes ABC and A'B'C' is 'd'.

$$\therefore d = d_2 - d_1 = \frac{1}{\sqrt{\frac{b^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}}} \quad (3.8)$$

Using Equation (3.8), we can determine the interplanar separation in orthorhombic crystals. For tetragonal crystal $a = b \neq c$, substitute these values in Equation (3.8), we have:

$$d = \frac{1}{\sqrt{\frac{h^2}{a^2} + \frac{k^2}{a^2} + \frac{l^2}{c^2}}} = \frac{1}{\sqrt{\frac{h^2 + k^2}{a^2} + \frac{l^2}{c^2}}} \quad (3.9)$$

For cubic crystals: $a = b = c$, substitute these values in Equation (3.8), we have:

$$d = \frac{1}{\sqrt{\frac{h^2}{a^2} + \frac{k^2}{a^2} + \frac{l^2}{a^2}}} \quad (\text{or}) \quad d = \frac{a}{\sqrt{h^2 + k^2 + l^2}} \quad (3.10)$$

The calculation of interplanar spacing for other crystal systems is complicated, so we will not discuss them.

3.3 Imperfections in crystals

In a sound crystal (or in an ideal crystal), the atoms are arranged regularly and periodically in three dimensions. But the grown crystals [or real crystals] may contain imperfections or defects. These defects are mainly divided into point, line, surface and volume defects. Point and line defects are discussed here, while surface defects are dealt with in Appendix B.

(1) Point defects: As the name indicates, these defects are at some points in the crystal. So, these are also called zero-dimensional defects. The point defects are divided into three categories: (a) lattice site defects; this includes vacancies [Schottky defect] and interstitialcies [Frenkel defect], (b) compositional defects; this includes substitutional impurity and interstitial impurity and (c) electronic defects. These defects are discussed below.

(a) Lattice site defects: In this type of defects, some atoms may not be present in their regular atomic sites. They are:

(i) Vacancies: As shown in Fig. 3.5, at a lattice point, one or two or three atoms are missed, and this is referred to as single or double or triple vacancies, respectively. The vacancies are formed due to the imperfect packing during crystallization or due to thermal vibrations at high temperatures.

Figure 3.5 (a) Perfect crystal; (b) Vacancy defect

